

Antiderivatives and Riemann Sums



Basic antiderivatives

- 1 Find the general antiderivative of each function:
 - a $f(x) = x^4$
 - b $f(x) = \frac{1}{x}$
 - c $f(x) = e^x$
 - d $f(x) = \sec^2 x$
 - 2 Find $\int (3x^2 - 4x + 5) dx$.
 - 3 Find the particular antiderivative $F(x)$ of $f(x) = 6x^2 - 4$ satisfying $F(1) = 5$.
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Riemann sums: left, right, and midpoint

- 4 Approximate $\int_0^4 x^2 dx$ using a Left Riemann Sum with $n = 4$ equal subintervals.
 - 5 Approximate $\int_0^4 x^2 dx$ using a Right Riemann Sum with $n = 4$ equal subintervals.
 - 6 Approximate $\int_0^4 x^2 dx$ using a Midpoint Riemann Sum with $n = 4$ equal subintervals.
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The Trapezoidal Rule

- 7 Use the Trapezoidal Rule with $n = 4$ to approximate $\int_0^4 x^2 dx$.
 - 8 A table gives $f(0) = 2$, $f(2) = 5$, $f(4) = 9$, $f(6) = 8$. Use the Trapezoidal Rule with these unequal-width data (widths of 2 each) to approximate $\int_0^6 f(x) dx$.
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Sums as limits: the definite integral

- 9 Explain how the definite integral $\int_a^b f(x) dx$ is defined as a limit of Riemann sums, and write the limit expression using sigma notation.
 - 10 Determine whether the Left Riemann Sum overestimates or underestimates $\int_0^4 x^2 dx$, and explain why using the shape of the graph.
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Riemann sums from a table of values

- 11 A table gives $f(0) = 3$, $f(1) = 5$, $f(2) = 8$, $f(3) = 10$, $f(4) = 9$. Use a Right Riemann Sum with 4 subintervals of width 1 to approximate $\int_0^4 f(x) dx$.
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Antiderivatives of trigonometric and exponential functions

- 12 Find $\int (3\sin x + 2e^x) dx$.
- 13 Find the particular antiderivative $F(x)$ of $f(x) = \sec^2 x$ satisfying $F(0) = 2$.